

# Exploratory Figures

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2026-03-18

```
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.4
## v forcats    1.0.0      v stringr   1.5.0
## v ggplot2    3.5.2      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## Warning: package 'here' was built under R version 4.3.3
## here() starts at /Users/alonsovelasquez/Documents/ENV710/FinalProject/CiraTrishaAlonso
cleaned_seaotter_data <- read_csv(here("Data/cleaned_seaotter_data.csv"))

## Rows: 18899 Columns: 11
## -- Column specification -----
## Delimiter: ","
## chr  (6): sex_cd, site_name, success_cd, preytype_cd, preysize_cd, phylum
## dbl  (4): year, month, dive_time, prey_qty
## date (1): bout_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Does longer bout time equal more successful forage capture?

```
cleaned_seaotter_data <- cleaned_seaotter_data %>%
  filter(success_cd == "y" | success_cd == "n") %>%
  filter(sex_cd != "u")

dive_time_summary <- cleaned_seaotter_data %>%
  group_by(success_cd) %>%
  summarize(
    mean_dive_time = mean(dive_time, na.rm = TRUE),
    sd_dive_time = sd(dive_time, na.rm = TRUE),
    sample_size = n()
```

```

) %>%
ungroup()

print(dive_time_summary)

## # A tibble: 2 x 4
##   success_cd mean_dive_time sd_dive_time sample_size
##   <chr>          <dbl>         <dbl>      <int>
## 1 n             63.0          40.6       1262
## 2 y             72.3          34.6      17325

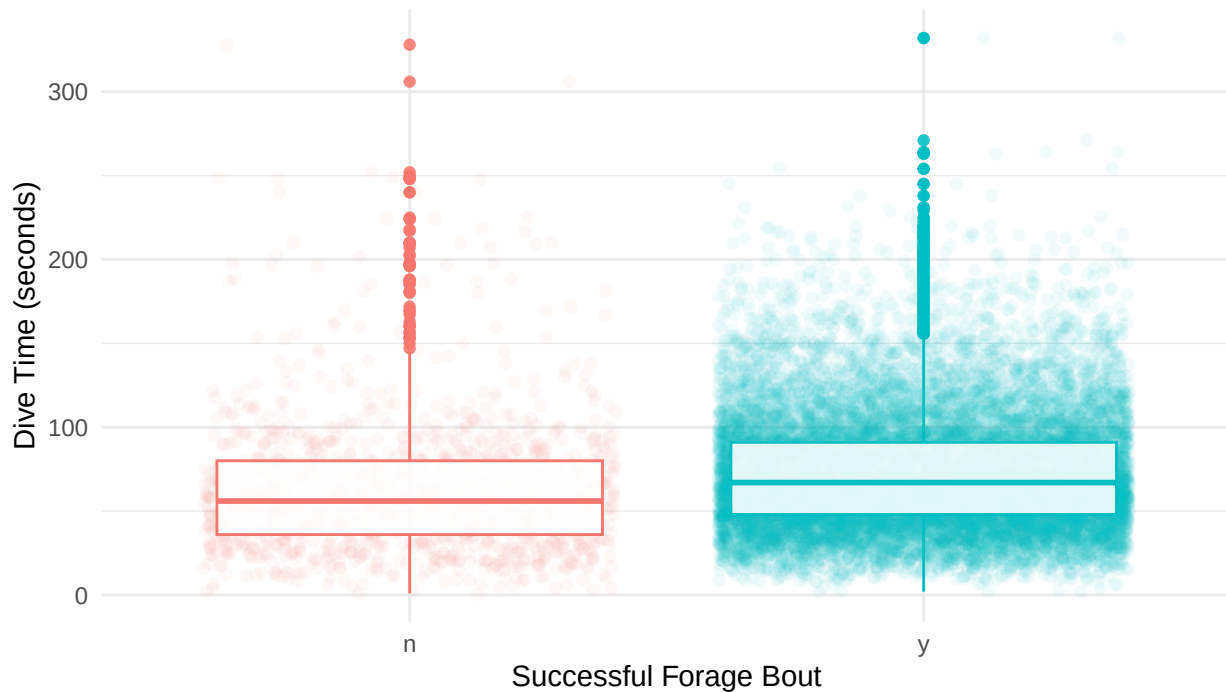
cleaned_seaotter_data %>%
  ggplot(aes(x= success_cd, y = dive_time, color = success_cd)) +
  geom_jitter(size = 1, alpha = 0.05, stroke = 1) +
  geom_boxplot(alpha= 0.88) +
  theme_minimal() +
  labs(x = "Successful Forage Bout", y = "Dive Time (seconds)", title = "Sea Otter Dive Time and Forage
  theme(legend.position = "none")

## Warning: Removed 868 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
## Warning: Removed 868 rows containing missing values or values outside the scale range
## (`geom_point()`).

```

### Sea Otter Dive Time and Forage Bout Success

Unsuccessful forage bouts had an average dive time of 63.0s (sd = 40.6), while successful forage bouts had an average dive time of 72.3s (sd = 34.6). There is a statistically significant difference in means between unsuccessful and successful forage bouts according to a two-tailed two-sample t-test ( $t = 7.85$ ,  $df = 1376.1$ ,  $p = 8.01 \times 10^{-16}$ ).



```

#Looking at variables
#examining dive time by gender
ggplot(cleaned_seaotter_data, aes(x=dive_time)) +

```

```
geom_histogram() +
labs(x = "Dive Time (seconds)",
     y = "Count", title = "Histogram of Sea Otter Dive Time per Forage Bout (Grouped by Sex)",
     subtitle = "Data seems to be normally distributed, with slight right tails") +
facet_grid(~sex_cd)
```

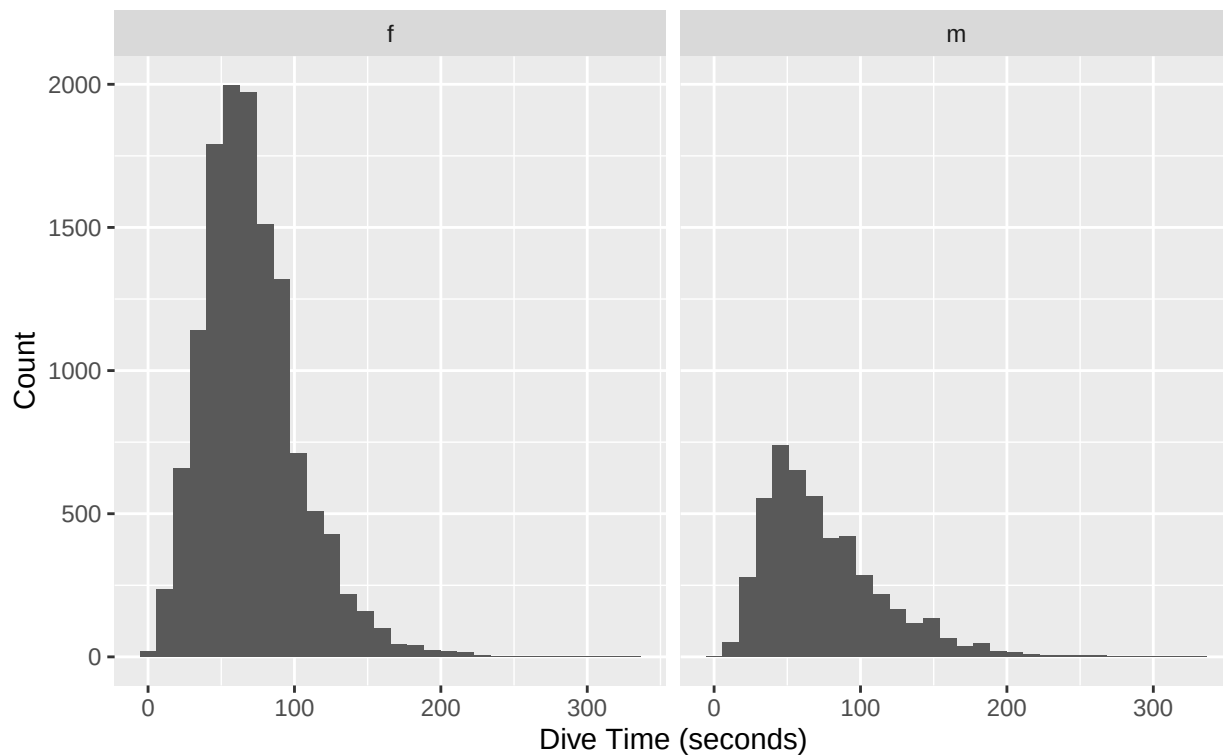
## `stat\_bin()` using `bins = 30`. Pick better value with `binwidth`.

## Warning: Removed 868 rows containing non-finite outside the scale range

## (`stat\_bin()`).

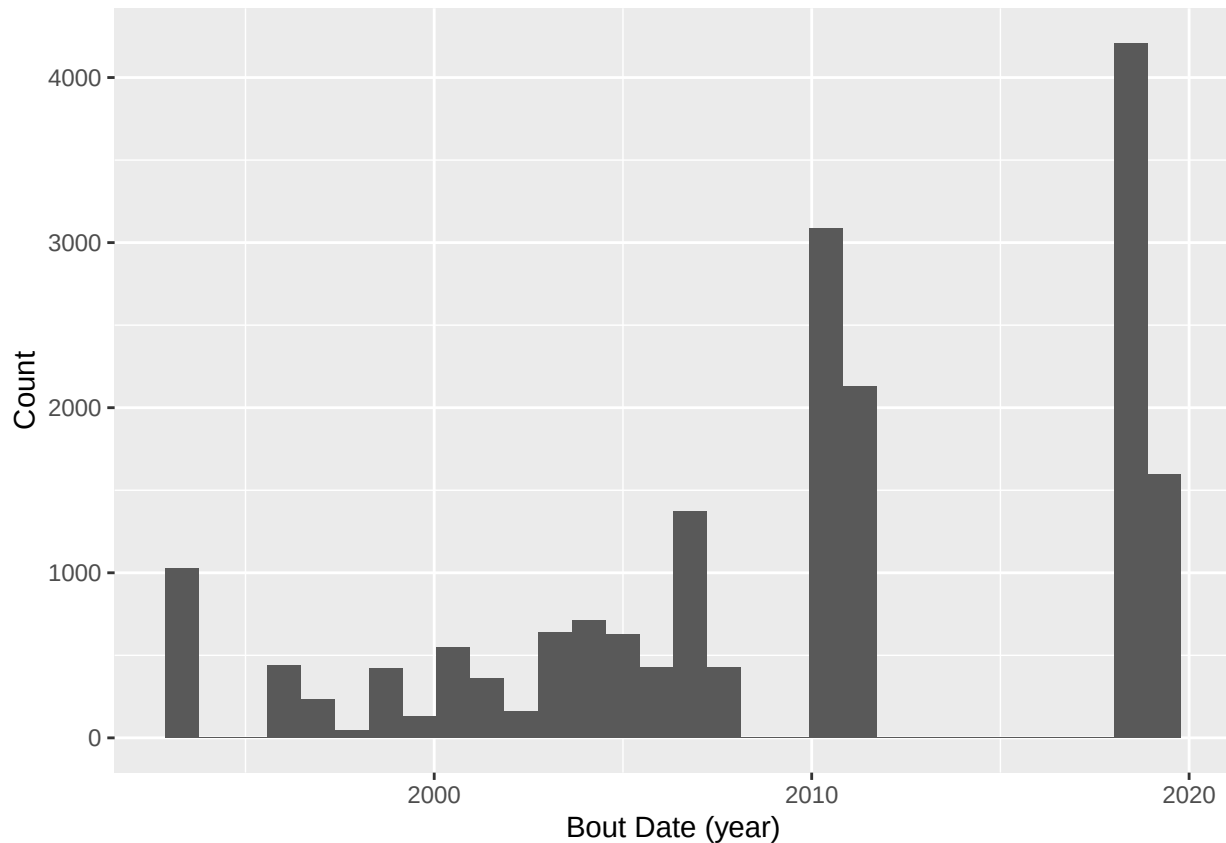
## Histogram of Sea Otter Dive Time per Forage Bout (Grouped by Sex)

Data seems to be normally distributed, with slight right tails



```
#examining bout date
ggplot(cleaned_seaotter_data, aes(x=bout_date)) +
  geom_histogram() +
  labs(x = "Bout Date (year)",
       y = "Count")
```

## `stat\_bin()` using `bins = 30`. Pick better value with `binwidth`.



```
#T-Test!!!
```

```
#creating two different datasets
```

```
successful_dive_time <- cleaned_seaotter_data %>%
```

```
  filter(success_cd == "y")
```

```
unsuccessful_dive_time <- cleaned_seaotter_data %>%
```

```
  filter(success_cd == "n")
```

```
successful_dive_ttest <- t.test(x = successful_dive_time$dive_time,
                                y = unsuccessful_dive_time$dive_time)
```

```
successful_dive_ttest
```

```
##
```

```
##  Welch Two Sample t-test
```

```
##
```

```
## data:  successful_dive_time$dive_time and unsuccessful_dive_time$dive_time
```

```
## t = 7.8548, df = 1376.1, p-value = 8.011e-15
```

```
## alternative hypothesis: true difference in means is not equal to 0
```

```
## 95 percent confidence interval:
```

```
##    6.985022 11.635367
```

```
## sample estimates:
```

```
## mean of x mean of y
```

```
##  72.30212  62.99193
```

```
#does success change by site name?
```

```

seaotter_success_site <- cleaned_seaotter_data %>%
  filter(success_cd == "y") %>%
  group_by(site_name) %>%
  count(success_cd) %>%
  rename(count_success = "n")

seaotter_failure_site <- cleaned_seaotter_data %>%
  filter(success_cd == "n") %>%
  group_by(site_name) %>%
  count(success_cd) %>%
  rename(count_failure = "n")

proportion_success_site <- seaotter_success_site %>%
  left_join(seaotter_failure_site %>%
    select(count_failure), by = "site_name")

```

```
## Adding missing grouping variables: `site_name`
```

```

proportion_success_site <- proportion_success_site %>%
  mutate(proportion_success = count_success / (count_success+count_failure))

```